

Internet Computing

CSS

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Structure and Appearance

- CSS (**C**ascading **S**tyle **S**heets) is a language for controlling the *style* of HTML elements
- In other words, it controls what things look like
- The idea is that the *content* of the things in the web page is separate from the *appearance* of the things
- CSS is also important for controlling things in the web page, we will look at that in other presentations

Most HTML Elements have a Default Style

- HTML elements already have some default visual style, for example:
 - `<b>` gives you bold text
 - `<h1>` gives you big size and bold text
- CSS allows you to further control the element style

Beautiful

I look **absolutely great!**



```
<h1>Beautiful</h1>
```

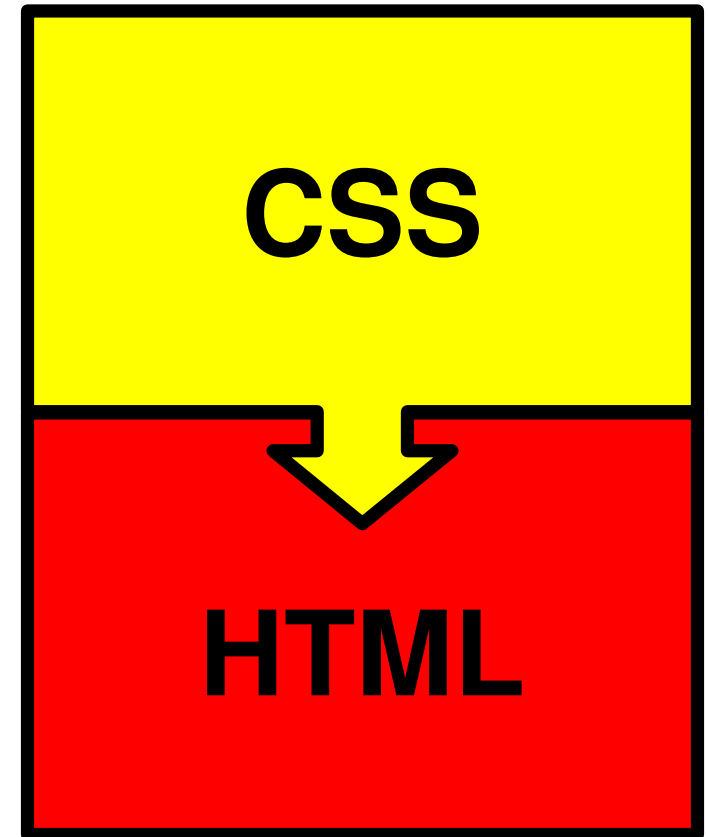
```
<p>
```

```
I look <b>absolutely  
great</b>!
```

```
</p>
```

Simple Way to Think of CSS

- The web page content is put in the HTML
- A set of *style rules* is put in the CSS, which is then applied to the HTML
- In this way the *content* is separate from the rules which control its *appearance*



An Example

HTML Content

```
<h1>Beautiful</h1>
<p>
  I look
  <b>absolutely great</b>!
</p>
```

CSS Rules

```
body { font-family: Arial;
        background: deeppink; }
h1    { color: darkred; }
p     { color: white; }
```

Beautiful

I look absolutely great!

What the page looks like

CSS Rules

```
body { font-family: Broadway;
        background: blue; }
h1    { color: white; }
p     { color: black; }
```

Beautiful

I look absolutely great!

What the page looks like

The Complete Page

```
<!DOCTYPE html>
<html>
<head>
  <title>Beautiful</title>
  <style>
    body { font-family: Arial;
           background: deeppink; }
    h1    { color: darkred; }
    p     { color: white; }
  </style>
</head>
```

- This is the complete code for the example on the left of the last slide

Beautiful

I look **absolutely great!**



```
<body>
  <h1>Beautiful</h1>
  <p>
    I look
    <b>absolutely great</b>!
  </p>
</body>
</html>
```

Style Rules

- A style rule has a selector and a collection of properties enclosed by a pair of braces:

h1 { color: darkblue; }

Selector *One or more CSS properties*

- The selector is a way to tell the HTML document which elements that the rule applies to
- In the example, the rule is applied to *every* `<h1>` element inside the document

Style Sheets

- Multiple style rules can be put in an HTML document inside `<style>...</style>`
- The collection of style rules is then called a *style sheet*, for example:

```
<style>
body { font-family: Arial;
      background: deeppink; }
h1   { color: darkred; }
p    { color: white; }
</style>
```



The style sheet of the previous example

- Style sheets are commonly put inside the head section of an HTML document

A Style Sheet in HTML

```
<html>
<head>
  <title>Using a Style Sheet</title>
  <style>
    h1 { color: darkblue; }
  </style>
</head>
<body>
  <h1>Name</h1>
  <p>David Lam</p>
  <h1>Job</h1>
  <p>Teacher</p>
  <h1>Hobbies</h1>
  <p>Surfing the web</p>
</body>
</html>
```



*A style sheet with
only one rule, which
changes all <h1>
text to darkblue*

Name

David Lam

Job

Teacher

Hobbies

Surfing the web

Spaces Between the Content Are Not Important

- `h1{color:darkblue;}`
 - `h1 { color: darkblue; }`
 - `h1 { color :darkblue;}`
 - `h1 { color: darkblue; }`
- These are all okay*
- `h1 { color:dark_blue; }`
- Not okay! You can't have a space in the middle*

The Semicolon is For Separating

- Previously we saw this:

```
body { font-family: Arial;  
      background: deeppink; }
```

- The semicolon ; is only needed for separation
- So you could do this:

```
body { font-family: Arial;  
      background: deeppink }
```

No need for semicolon at the end



Linking to External Style Sheets

- If you want to, you can put the style sheet in a separate file and then link it back using the `<link>` tag, like this:

```
<link href="mystyle.css" rel="stylesheet">
```



*This file contains the style sheet
(in the same folder as
the HTML file in this example)*

Using an External Style Sheet

```
<html>
<head>
  <title>Using an External Style Sheet</title>
  <link href="mystyle.css" rel="stylesheet">
</head>
<body>
  <h1>Name</h1>
  <p>David Lam</p>
  <h1>Job</h1>
  <p>Teacher</p>
  <h1>Hobbies</h1>
  <p>Surfing the web</p>
</body>
</html>
```

```
h1 { color: darkblue; }
```

mystyle.css

*The visual result
is exactly the same*

Name

David Lam

Job

Teacher

Hobbies

Surfing the web

Inline Style

- One way to apply style to one single thing is this:

Beautiful

I look **absolutely** great!

```
<h1 style="color:red">Beautiful</h1>
```

- This means the text of that <h1> becomes red
- Everything else in the web page is unaffected
- This way of applying style is called *inline style*

There Are Many CSS Properties

- There are hundreds of CSS properties
- You can find a list on the web:

`https://www.w3.org/Style/CSS/
all-properties.en.html`

- There are many CSS properties that you will probably never use!
- In this presentation we look at some of the more common CSS properties

Examples of Some Common Font Properties

```
<p style="font-family: Helvetica;  
        font-weight: bold;  
        font-size: 18px">
```

Big Big Text</p>

```
<p>normal text</p>
```



Big Big Text

normal text

Common Font Properties

- `font-family: Helvetica`
 - Other examples: "Times New Roman", Arial, Georgia, etc

You need to enclose the font name with double quotes when it contains spaces

- `font-weight: bold`
 - Other examples: normal, bolder, lighter
 - It can also be a number from 100 (very thin), 200, 300, ... to 1000 (very thick)

Common Font Properties

- `font-size: 18px`
 - Other examples: 8px, 40px, 90px, etc
 - There are lots of different ways to express font size
 - In this example, we used pixels (px), but you can also use points (pt), relative size to the element (em), relative size to the document (rem), and so on – we probably won't explore all these

Common Colour Properties

- `background-color: lightgray`
 - Or you can use `background`, i.e.
`background: lightgray`
- `color: darkgray`
 - This is the foreground (i.e. text) colour
- There are different ways to specify colours, for example, using colour names, RGB colours or HSL colours (see next example)

The HTML Span Tag

- `<span>...</span>` is useful because it doesn't have any default visual style, e.g.:

`<p>In a paragraph <span>In a span</span></p>`

In a paragraph In a span

*No extra style has been
used by the `<span>`*

- So we can apply styles to `<span>...</span>` without worrying whether it already has a style which might mess up the result in some way

Examples of Controlling Colour



RAINBOW

```
<body style="background-color: skyblue">  
  <p>  
    <span style="color: red">R</span>  
    <span style="color: orange">A</span>  
    <span style="color: yellow">I</span>  
    <span style="color: green">N</span>  
    <span style="color: #00F">B</span>  
    <span style="color: #400080">O</span>  
    <span style="color: rgb(255, 0, 255)">W</span>  
  </p>  
</body>
```

Different ways of using colours, see next slide

Specifying Colours

- Colours can be names, i.e.:

color: yellow { *Use common colour names*

- Or short RGB hexadecimal values:

color: #FF0 { *Each of red, green and blue is a value from 0 to F*

- Or long RGB hexadecimal values:

color: #FFFFFF00 { *Each of red, green and blue is a value from 00 to FF*

- Or decimal RGB values:

color: rgb(255,255,0) { *Each of red, green and blue is a value from 0 to 255*

Common Paragraph Properties

- `text-align: center`
 - Other examples: `left`, `right`, `justify`
- `line-height: 1.6`
 - This increases the vertical gap between the lines

```
<p style="text-align: center;  
        line-height: 1.6">
```



*The result of applying
these rules is shown
on the next slide*

Good day!

This text has a nice gap between the lines.

That makes it easier to read.

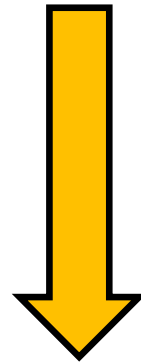
But you won't see it unless there lots of text,
or if the browser is not very wide.

```
</p>
```

Examples of Paragraph Properties

Good day! This text has a nice gap between the lines.
That makes it easier to read. But you won't see it unless
there lots of text, or if the browser is not very wide.

*The default appearance
of the paragraph*



Apply these rules:

```
text-align: center;  
line-height: 1.6
```

Good day! This text has a nice gap between the lines.
That makes it easier to read. But you won't see it unless
there lots of text, or if the browser is not very wide.

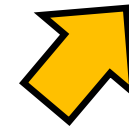
Inline and Block Elements

- There are generally two types of layout:
 1. Elements that ‘flow’ with the page are called *inline elements*, such as
`<b>...</b>` and `<span>...</span>`
 2. Elements that occupy a rectangular area on its own line are called block elements, such as
`<h1>...</h1>` and `<p>...</p>`
 - These start on a new line

Examples of Box Properties

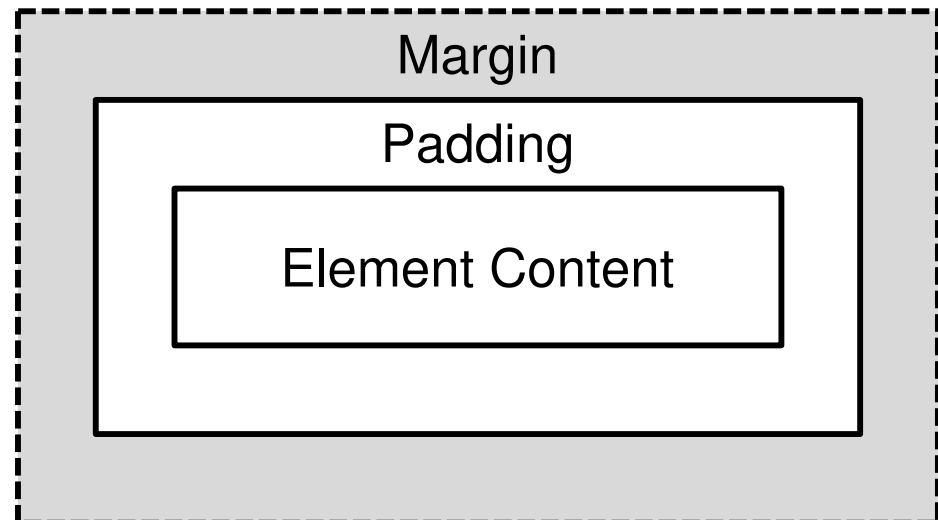
In a Big Box

```
<h1 style="background: yellow;  
margin: 20px;  
padding: 20px;  
width: 200px; height: 40px">
```

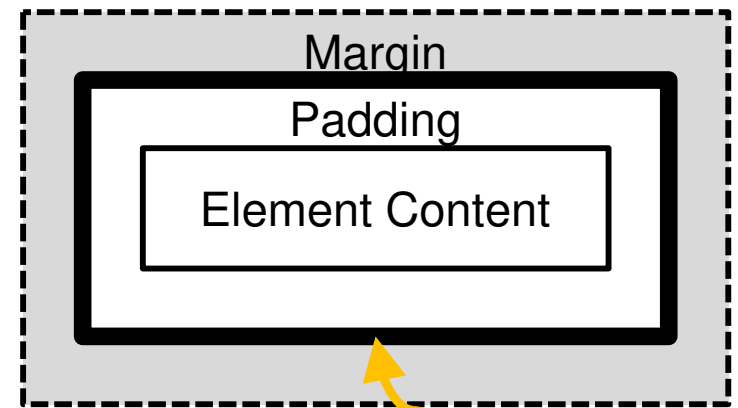


In a Big Box
</h1>

*The 'box'
around an
element*



Border Properties



- `border: 2px solid black`
thickness of the line *line style* *colour*
The border of an element
- Line style can also be dotted, dashed, double etc
- `border-radius: 10px`
 - This makes the corners of the thing which has the border using a curve

Examples of Border Properties

```
<h1 style="background: red;  
        width: 180px;  
        text-align: center;  
        border: 2px solid black;  
        border-radius: 20px">
```

Warning!

```
</h1>
```

*The border
is curved at
the corners*



The HTML Div Element

- You have seen the use of `<span>...</span>`
- `<div>...</div>` is another HTML element without any default visual style
- It is a block element, which means it starts on a new line
- `<span>...</span>` and `<div>...</div>` are very useful if you want to build your own page style

Examining CSS

- It is useful to look at websites (your web site or another) and study their style using the Chrome Developer tools

Internet Computing >
CSS Examples

Move mouse on something, right click, then select 'Inspect'

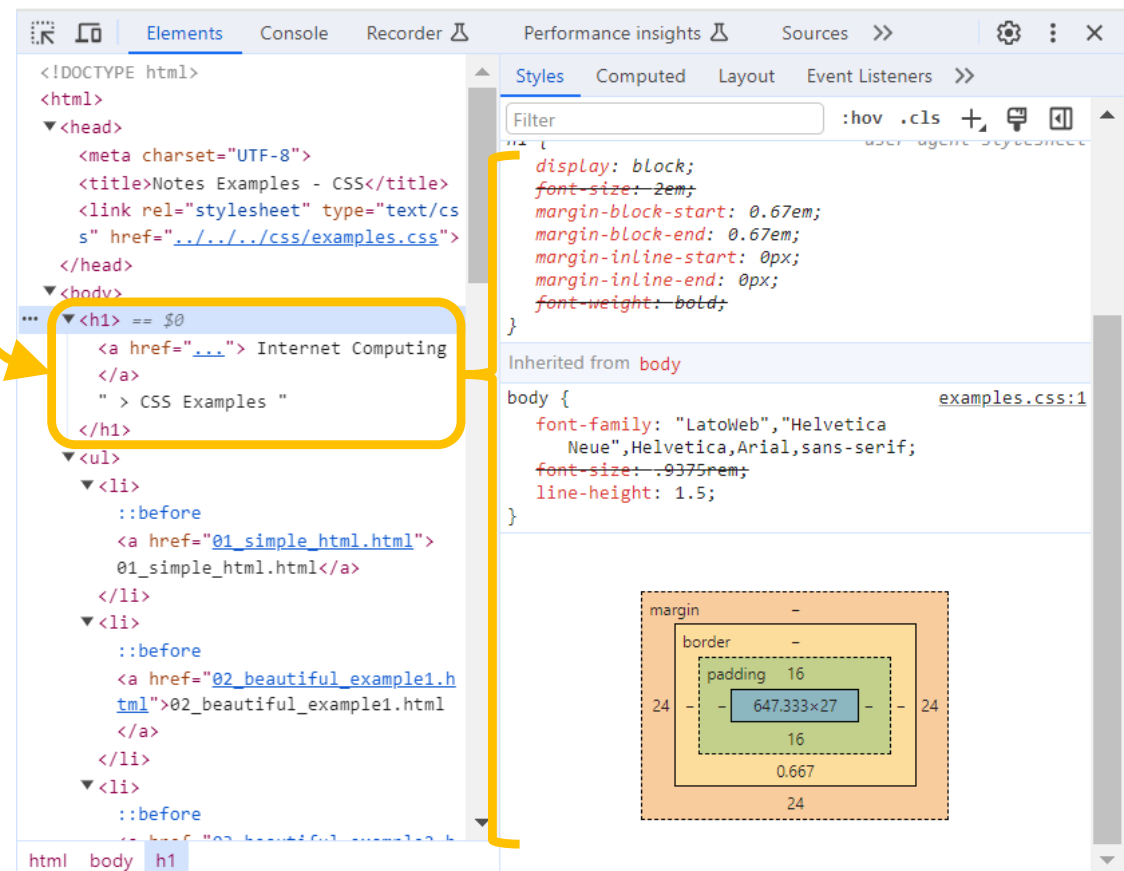
Or use shortcut keys:



Ctrl+Shift+i



⌘+Shift+c



The id Attribute

- You can give a 'name' to any HTML using id, for example:

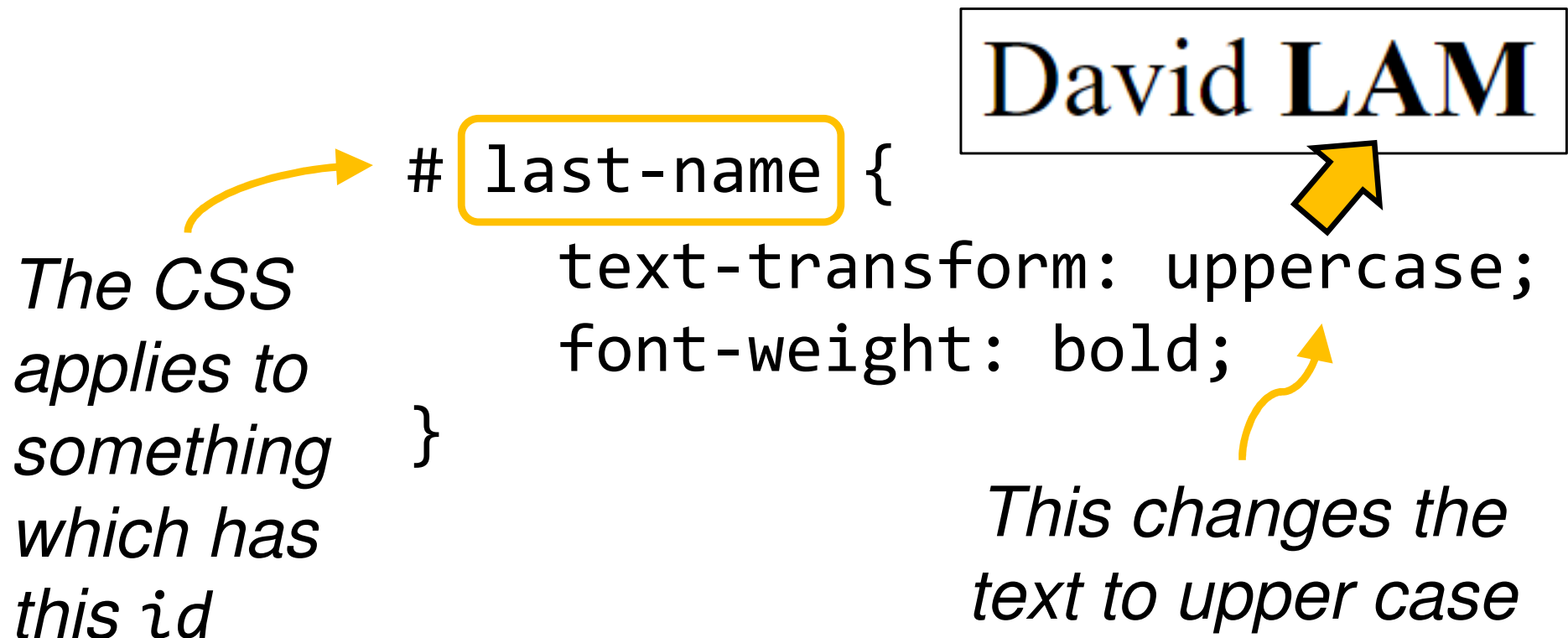


```
<p>  
    <span id="first-name">David</span>  
    <span id="last-name">Lam</span>  
</p>
```

- Then you can make a style rule which applies to only the thing with that id – see next slide
- Giving things an id is also very useful for programming

Using the id

- This is how you make a rule which applies to the thing which have that id



The Class Attribute

- Using `class` is another way to apply a style rule to some HTML, for example:

`<p>The meeting next week is on`

`<span class="highlight">Tuesday</span>`

`starting from`

`<span class="highlight">9am to 10am</span>.`

`</p>`

These two spans are in the same class so the same rule can be applied to them – see next slide

Example of Using Class

The rule applies to the things with the class

```
.highlight {  
    background: yellow;  
}
```

The meeting next week is on Tuesday
starting from 9am to 10am.

Quick Summary of Using CSS

- Three ways to use style rules:

Example HTML

CSS Rule

`<p>My text</p>`

`p { color: blue }`

`<p id="mytext">My text</p>`

`#mytext { color: blue }`

`<p class="fun">My text</p>`

`.fun { color: blue }`